WEEKLY BRIEF · 2026-11-02

机器人学研究进展周报

Robotics Research & Industry · Weekly Brief

北京时间 2026-11-02 发布 · 5 条进展

AI生成：内容由 AI 辅助整理与总结，请以原始来源为准。

覆盖北京时间 2026-10-26 00:00 至 2026-11-01 24:00 首次公开的信息。

本周重点

ANYbotics 宣布开始向客户交付防爆巡检足式机器人 ANYmal X

ANYbotics announces first customer deliveries of ANYmal X

机构宣称，未独立验证 · 原文日期 2026-10-27

ANYbotics 10 月 27 日宣布开始向包括 Equinor 和 DSM-Firmenich 在内的客户交付 ANYmal X，用于油气和化工设施的危险区域巡检。公司称该系统采用热成像、视觉、声学和气体传感器，并接入其 Shift 运维平台。

ANYbotics said it began shipping ANYmal X to customers including Equinor and DSM-Firmenich for hazardous-area inspections in oil, gas and chemical plants. It describes thermal, visual, acoustic and gas sensing integrated with its Shift platform.

意义：防爆环境对移动机器人硬件与认证有特殊要求，客户交付是从早期测试走向实际应用的新节点。

局限：交付数量、客户现场运行结果和独立认证文件未在该公告中提供；“ 全球首款 ”仅为公司表述。

原始来源：https://www.anybotics.com/news/anymal-x-launch/

GXO 与 Agility 宣布将 Digit 仓储部署扩展到美国两处站点

GXO and Agility announce expansion of Digit deployment to two US sites

机构宣称，未独立验证 · 原文日期 2026-10-26

GXO 与 Agility 10 月 26 日宣布扩大合作，计划于 2027 年初在北卡罗来纳州 Statesville 和田纳西州 Lebanon 的 GXO 站点部署 Digit 4，从事重复性物料搬运，并在 Digit 5 商业化后迁移。原有 Georgia 站点已运行，但新站点尚未开始部署。

GXO and Agility announced plans to start Digit 4 deployments in early 2027 at GXO sites in Statesville, North Carolina, and Lebanon, Tennessee, for repetitive material handling, with a later migration to Digit 5. The new sites are planned, while the existing Georgia site is operating.

意义：客户与供应商联合公告提供了具体站点和预期部署时间，可追踪人形机器人多站点复制的后续结果。

局限：新站点部署属于未来计划，未披露台数或验收指标；双方未提供独立性能测试。

原始来源：https://www.agilityrobotics.com/content/gxo-and-agility-robotics-expand-partnership-to-accelerate-deployment-of-humanoids-in-u-s-operations

学术研究

单一强化学习策略在变动飞行条件下控制吊载无人机的性能表征

Characterization of a fixed reinforcement learning policy for aerial robot with suspended payload under variable flight conditions

同行评审 · 原文日期 2026-10-26

第一作者：Ali Tahir Karasahin
第一作者单位：School of Civil, Aerospace and Design Engineering, University of Bristol, Bristol, BS8 1QU, UK；Faculty of Engineering, Department of Mechatronics Engineering, Necmettin Erbakan University, 42090, Konya, Turkey；Bristol Robotics Laboratory, University of the West of England (UWE), Bristol BS16 1QY, UK

论文摘要原文 / Original abstract

Flights with suspended payloads are particularly challenging because of their coupled dynamics, which lead to instability and increased sensitivity to disturbances. Although reinforcement learning (RL) has successfully achieved controller performance, the generalization and robustness of a single policy remain significant areas of investigation. In this study, we characterized the performance and robustness of a single RL policy for an aerial robot with different trajectory profiles, including a smooth, feasible lemniscate curve and a sharp-turning, infeasible pentagram, under varying velocity references (0.5 m s–1 and 1.0 m s–1) and crosswind disturbances (1.0 m s–1). We trained a single RL policy using proximal policy optimization with collective thrust and body-rate control using a high-fidelity physics simulator based on the SimpleFlight framework. Real-world experimental results on the Crazyflie 2.1 platform show that the single RL policy successfully generalizes to different trajectory profiles and velocity references and maintains stability under a crosswind disturbance of up to 1.0 m s–1 which is a substantial challenge for this small class platform and even smaller payload underneath, where such aerodynamic forces are significant compared to the available control authority and system mass. Furthermore, the single RL policy was systematically evaluated using the mean Euclidean distance error, cable length transitions, and swing angle distributions. Although the single RL policy maintained a robust control performance, the experimental results indicated performance degradation at higher velocities owing to increased dynamic challenges such as nonlinear aerodynamic drag and actuator saturation. This study provides a detailed performance characterization that highlights the generalization capability of a single-payload-aware RL policy in real-world applications and the limitations arising from the hybrid dynamics of the system.

摘要中文翻译

吊载飞行因耦合动力学而格外困难，这会导致不稳定并提高系统对扰动的敏感性。尽管强化学习（RL）已取得控制器性能方面的成果，单一策略的泛化性和鲁棒性仍是重要研究问题。本研究表征了单一强化学习策略在不同轨迹形状下控制吊载无人机的性能与鲁棒性：轨迹包括平滑且可行的双纽线，以及急转弯且不可行的五角星；速度参考值为 0.5 m s − 1 和 1.0 m s − 1，侧风扰动为 1.0 m s − 1。我们基于 SimpleFlight
框架的高保真物理仿真器，采用近端策略优化训练了一个以总推力和机体角速度为控制量的强化学习策略。在 Crazyflie 2.1 平台上的实机实验表明，该单一策略能够泛化到不同轨迹和速度参考值，并在最高 1.0 m s − 1 的侧风扰动下保持稳定；对于这种小型平台及其下方更小的吊载而言，相对于可用控制能力和系统质量，这类气动力是相当大的挑战。此外，研究系统评估了该策略的平均欧氏距离误差、绳长变化和摆角分布。虽然该策略保持了稳健的控制性能，实验结果显示，由于非线性气动阻力和执行器饱和等动态挑战增加，较高速度下性能下降。本研究详细表征了单一吊载感知强化学习策略在真实应用中的泛化能力，以及系统混合动力学带来的局限。

摘要来源：Karasahin, A. T., Wu, Z. & Kocer, B. B., Scientific Reports 16, 34215 (2026), CC BY 4.0。中文译文由 RoboticsNews 翻译；数学单位改为纯文本排版。

意义：对单策略跨工况适用范围给出实机表征，可供吊载飞行控制研究参考。

局限：实验平台为 Crazyflie 2.1；论文指出较高速度下性能下降，结论不应外推到其他机型或风况。

原始来源：https://www.nature.com/articles/s41598-026-46111-x
摘要原文：https://www.nature.com/articles/s41598-026-46111-x
单位依据：https://www.nature.com/articles/s41598-026-46111-x
摘要许可 CC BY 4.0；中文译文由 RoboticsNews 翻译：https://creativecommons.org/licenses/by/4.0/

生态动态

Agility 与 Churchill XI 预计合并相关融资总额约 4.65 亿美元

Agility and Churchill XI expect about $465 million in gross transaction financing

机构宣称，未独立验证 · 原文日期 2026-10-30

两家公司 10 月 30 日宣布，在赎回期限届满后，拟议业务合并及相关融资的预计总额约为 4.65 亿美元，尚未扣除交易费用。交易预计于 11 月 4 日完成，仍须 11 月 2 日股东表决及其他条件满足；并非已完成融资或上市。

The companies said expected gross financing tied to their proposed combination is about $465 million before transaction expenses after the redemption deadline. Closing is expected on November 4, subject to a November 2 shareholder vote and other conditions; financing and listing were not yet completed.

意义：若交易完成，将影响人形机器人公司的资本获取路径；目前应跟踪表决和实际交割。

局限：金额和完成日期均为公司预期，取决于交易条件；该数字不是当期收入。

原始来源：https://www.sec.gov/Archives/edgar/data/2074973/000121390026114063/ea031030301ex99-1.htm

Exotec 发布美加仓储自动化采购者调查：27% 称采购时运营尚未准备好

Exotec releases US–Canada warehouse automation buyer survey

机构宣称，未独立验证 · 原文日期 2026-10-28

Exotec 10 月 28 日发布一项覆盖美国和加拿大 250 名供应链决策者的调查。公司称超过四分之一受访者表示采购时业务尚未准备好，94% 认为更换供应商成本过高。结果反映受访者主观判断，而非所有仓库的实际绩效。

Exotec released a survey of 250 supply-chain decision-makers in the US and Canada. It reported that 27% said their operation was not ready when they invested and 94% considered switching providers prohibitively expensive. These are respondent views, not measured outcomes across all warehouses.

意义：采购准备度、系统转换成本与自动化部署决策直接相关，调查可作为后续独立研究的问题线索。

局限：该调查由仓储机器人供应商发布；样本仅限美加受访者，不能外推为全球仓储行业比例。

原始来源：https://www.exotec.com/news/one-in-four-supply-chain-leaders-regret-their-warehouse-automation-decision-and-94-say-switching-would-be-prohibitively-expensive/